Northwind Renewable Energy

Annual Operations & Sustainability Report

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1. Executive Summary

Northwind Renewable Energy operates a portfolio of wind, solar, and battery storage assets across four regions of North America. In fiscal year 2025, the company generated 4.2 terawatt-hours of clean electricity, a 14 percent increase over the prior year. Revenue reached 312 million dollars, driven primarily by expansion of the Cedar Ridge wind farm and the commissioning of the Blue Mesa solar array. Operating costs rose 9 percent due to higher maintenance spending on aging turbine components at the Fairhaven site.

The company's board approved a five-year capital plan totaling 1.1 billion dollars, with the largest allocation directed toward grid-scale battery storage. Management expects storage capacity to triple by fiscal year 2028, reducing curtailment losses that historically affected the Cedar Ridge and Prairie Junction facilities during periods of low demand.

> *“Fiscal year 2025 was a turning point for how we think about storage,” said Chief Executive Officer Elena Marsh in the shareholder letter. “We are no longer just generating power; we are learning to hold onto it and release it exactly when the grid needs it most.”*

2. Company Overview and History

Northwind Renewable Energy was founded in 2009 as a small wind development firm in Minnesota. The company's first commercial project, the 40-megawatt Willow Creek wind farm, entered service in 2011. Over the following decade, Northwind expanded through a combination of organic development and three acquisitions: SolarPeak Systems in 2016, Coastal Ridge Energy in 2019, and Sundown Battery Technologies in 2023.

The acquisition of Sundown Battery Technologies marked the company's formal entry into grid-scale storage. At the time of acquisition, Sundown operated a single 20-megawatt pilot facility; that facility has since been expanded to 120 megawatts and rebranded as the Sundown Battery Storage site within the Southwest region.

2.1 Corporate Timeline

Year	Milestone
2009	Company founded in Minnesota as a wind development firm.
2011	Willow Creek wind farm (40 MW) enters commercial service.
2016	Acquisition of SolarPeak Systems; company enters solar development.
2019	Acquisition of Coastal Ridge Energy; Pacific region operations begin.
2021	Cedar Ridge wind farm reaches 300 MW after phase two expansion.
2023	Acquisition of Sundown Battery Technologies.
2025	Blue Mesa solar array and expanded Sundown storage reach full operation.

Table 1: Corporate milestones, 2009 to 2025.

3. Regional Performance

Performance varied significantly across regions. The Midwest region, anchored by Cedar Ridge and Prairie Junction, contributed the largest share of generation, while the Southwest region posted the highest growth rate due to the Blue Mesa solar array reaching full commercial operation in March.

Region	Facility	Capacity (MW)	FY2025 Output (GWh)	Capacity Factor
Midwest	Cedar Ridge Wind	480	1,540	36.6%
Midwest	Prairie Junction Wind	310	980	36.1%
Southwest	Blue Mesa Solar	250	620	28.3%
Southwest	Sundown Battery Storage	120	N/A	N/A
Northeast	Fairhaven Wind	190	540	32.5%
Northeast	Harborlight Solar	95	210	25.2%
Pacific	Coastal Ridge Wind	275	890	36.9%
Pacific	Willow Creek Wind	40	118	33.7%

Table 2: Facility-level generation output and capacity factor for FY2025.

Fairhaven's below-average capacity factor of 32.5 percent was attributed to an extended maintenance outage in the second quarter, when six of thirty-one turbines were offline for gearbox replacement. Management does not expect similar downtime in fiscal year 2026. Harborlight Solar, the smallest facility in the portfolio, underperformed its design capacity factor of 29 percent due to two winter storms that damaged tracking motors on 40 panel rows in January.

3.1 Regional Revenue Contribution

Region	FY2024 Revenue ($M)	FY2025 Revenue ($M)	Change
Midwest	148	162	+9.5%
Southwest	41	68	+65.9%
Northeast	39	42	+7.7%
Pacific	51	40	-21.6%

Table 3: Regional revenue by fiscal year. Pacific region revenue declined due to a scheduled power purchase agreement repricing at Coastal Ridge that took effect in October 2024.

4. Financial Results

Total revenue for fiscal year 2025 was 312 million dollars, compared to 279 million dollars in fiscal year 2024. Net income was 38 million dollars, down from 44 million dollars in the prior year, reflecting higher depreciation expense from newly commissioned assets and one-time costs associated with the Fairhaven gearbox replacements.

Line Item	FY2024 ($M)	FY2025 ($M)
Revenue	279	312
Operating expenses	164	179
Depreciation & amortization	51	61
Interest expense	12	15
Net income	44	38
Capital expenditures	210	265

Table 4: Condensed income statement, FY2024 vs FY2025.

The company's debt-to-equity ratio increased from 0.61 to 0.74 during the year, reflecting new project financing for the Blue Mesa solar array and the Sundown storage expansion. Management has stated a long-term target of keeping the ratio below 0.85.

5. Safety and Workforce

The company recorded a total recordable incident rate of 0.8 per 200,000 hours worked, below the industry benchmark of 1.4. Workforce headcount grew to 1,240 employees, including 85 new hires in field technician roles to support the expanding storage fleet.

Key workforce initiatives launched in fiscal year 2025 included:

- A technician apprenticeship program in partnership with two regional community colleges.
- Expanded remote monitoring training to reduce unnecessary site visits.
- A revised heat-stress protocol for Southwest region field crews.
- A mentorship pairing between senior engineers and first-year hires.
- A new fall-protection harness standard rolled out across all wind sites.

5.1 Headcount by Region

Region	FY2024 Headcount	FY2025 Headcount
Midwest	412	455
Southwest	180	260
Northeast	265	270

Pacific	260	255
Corporate	38	40

Table 5: Employee headcount by region, FY2024 vs FY2025.

6. Environmental Impact

Northwind estimates that its fiscal year 2025 generation displaced approximately 1.9 million metric tons of carbon dioxide equivalent, based on regional grid emission factors published by government agencies in each operating jurisdiction. The company's own operational footprint, including vehicle fleets and site offices, totaled 8,400 metric tons of carbon dioxide equivalent, a 6 percent reduction from the prior year following the rollout of electric service trucks at the Cedar Ridge and Prairie Junction sites.

Water usage across all sites was minimal, limited primarily to panel washing at the Blue Mesa and Harborlight solar arrays. The company reports 4.1 million gallons of water used for panel cleaning in fiscal year 2025, all sourced from non-potable municipal supply.

6.1 Wildlife and Land Use

All wind facilities operate under site-specific avian and bat monitoring plans. Cedar Ridge recorded 14 documented avian mortalities in fiscal year 2025, consistent with prior-year averages and within the range anticipated in the facility's environmental permit. The company continues to fund a multi-year raptor migration study near the Prairie Junction site in cooperation with a regional wildlife agency.

7. Technology and Innovation

In fiscal year 2025, Northwind deployed a predictive maintenance platform across its wind fleet, using vibration and temperature sensors to flag gearbox and bearing issues before failure. Early results at the Coastal Ridge site suggest the system can identify developing faults an average of 23 days before they would previously have been detected through routine inspection.

The company also began piloting a machine learning forecasting model to improve day-ahead energy trading decisions. The model incorporates weather forecasts, historical output, and grid demand data. During its first two quarters of use at Blue Mesa, the model reduced forecast error by 11 percent compared to the company's previous statistical approach.

7.1 Battery Storage Technology

The Sundown Battery Storage facility uses lithium iron phosphate cells housed in 42 containerized units, each rated at approximately 2.9 megawatts and 11.6 megawatt-hours. The facility is configured to provide both energy arbitrage, charging during low-price periods and discharging during high-price periods, and frequency regulation services to the regional grid operator.

8. Risks and Outlook

Management identified three primary risks for fiscal year 2026: transmission interconnection delays affecting the Prairie Junction expansion, potential tariff changes on imported battery cells, and continued supply constraints for large turbine gearboxes. The company has diversified its battery cell suppliers across two additional vendors to mitigate the second risk.

Guidance for fiscal year 2026 projects generation of 4.8 to 5.1 terawatt-hours and revenue between 340 and 365 million dollars, assuming no material delay in the Prairie Junction interconnection. Management noted that a delay beyond the second quarter of fiscal year 2026 could push approximately 60 million dollars of expected revenue into fiscal year 2027.

8.1 Risk Summary Table

Risk	Likelihood	Potential Impact	Mitigation
Interconnection delay (Prairie Junction)	Medium	High	Early engagement with grid operator; phased commissioning plan.
Battery cell tariff changes	Medium	Medium	Diversified supplier base across three countries.
Turbine gearbox supply constraints	High	Medium	Increased spare parts inventory; predictive maintenance program.
Severe weather damage	Medium	Medium	Insurance coverage; reinforced tracking motors at solar sites.

Table 6: Principal risks identified for fiscal year 2026.

9. Governance

Northwind's board of directors consists of nine members, seven of whom are independent. The board's audit committee is chaired by an independent director with a background in utility-sector finance. In fiscal year 2025, the board added a new committee focused specifically on climate risk and storage strategy oversight, reflecting the growing share of capital allocated to battery projects.

Executive compensation includes a performance component tied to safety metrics and generation targets. In fiscal year 2025, 30 percent of the executive team's variable compensation was linked to the total recordable incident rate remaining below 1.0.

10. Frequently Asked Questions

Q: What is Northwind's largest facility by capacity?
A: Cedar Ridge Wind, located in the Midwest region, is the largest facility at 480 megawatts of capacity.

Q: Which facility had the highest capacity factor in fiscal year 2025?
A: Coastal Ridge Wind in the Pacific region had the highest capacity factor at 36.9 percent.

Q: Why did Pacific region revenue decline in fiscal year 2025?
A: A scheduled power purchase agreement repricing at the Coastal Ridge facility took effect in October 2024, reducing the average price received for electricity sold from that site.

Q: How much battery storage capacity does Northwind operate?
A: The Sundown Battery Storage facility provides 120 megawatts of storage capacity, and management expects total storage capacity to triple by fiscal year 2028.

Q: What caused the underperformance at Harborlight Solar?
A: Two winter storms in January damaged tracking motors on 40 panel rows, reducing the facility's capacity factor below its design target of 29 percent.

Q: Who is Northwind's Chief Executive Officer?
A: Elena Marsh serves as Chief Executive Officer and is quoted in the shareholder letter included in the executive summary of this report.

11. Glossary

Capacity Factor — The ratio of actual energy output over a period to the maximum possible output at full capacity.

Curtailment — The deliberate reduction of power output below what could otherwise be produced, usually due to grid constraints.

Interconnection — The point and process by which a power generation facility connects to the electrical grid.

MW / GWh — Megawatts measure capacity; gigawatt-hours measure energy actually produced over time.

Frequency Regulation — A grid service in which a resource rapidly adjusts output to help maintain the grid's target operating frequency.

Power Purchase Agreement (PPA) — A long-term contract under which a generator sells electricity to a buyer at an agreed price.

Lithium Iron Phosphate (LFP) — A battery chemistry known for thermal stability and long cycle life, used in the Sundown storage facility.

Total Recordable Incident Rate (TRIR) — A standard safety metric representing the number of recordable injuries per 200,000 hours worked.

Appendix A: Facility Directory

The table below lists all operating facilities as of the end of fiscal year 2025, including commissioning year and primary technology type.

Facility	Region	Technology	Commissioned	Capacity (MW)
Willow Creek Wind	Pacific	Wind	2011	40
Cedar Ridge Wind	Midwest	Wind	2014 (exp. 2021)	480
Prairie Junction Wind	Midwest	Wind	2017	310
Coastal Ridge Wind	Pacific	Wind	2019	275
Fairhaven Wind	Northeast	Wind	2015	190
Harborlight Solar	Northeast	Solar	2020	95
Blue Mesa Solar	Southwest	Solar	2025	250
Sundown Battery Storage	Southwest	Battery Storage	2023 (exp. 2025)	120

Table 7: Full facility directory with commissioning years.